**Subject:** Mathematics
**Class:** Class X (CBSE)

**CBSE CLASS 10 – PERIODIC TEST 07**
**Chapter 1: Real Numbers**

**Time:** 60 Minutes
**MM:** 30

**Student Name:** _____ **Roll No.:** _____ **Date:** _____

**GENERAL INSTRUCTIONS:**

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

## SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

**Q1.** For any positive integer $n$, the expression $n^3 - n$ is always divisible by: [1]
(a) 6 (b) 9 (c) 12 (d) 15

**Q2.** If the sum of the LCM and HCF of two numbers is 1260 and their LCM is 900 more than their HCF, then the product of the two numbers is: [1]
(a) 194400 (b) 203400 (c) 184400 (d) 190000

**Q3.** If $p$ is a prime number, then $\sqrt{p}$ is: [1]
(a) a rational number (b) a natural number
(c) an irrational number (d) an integer

**Q4.** The number $3.24636363\ldots$ is: [1]
(a) an integer (b) a rational number
(c) an irrational number (d) a natural number

**Q5.** The number $7 \times 11 \times 13 \times 15 + 15$ is: [1]
(a) a prime number (b) a composite number
(c) an irrational number (d) neither prime nor composite

**Q6.** **Assertion (A):** The number $7 \times 11 \times 13 + 13$ is a composite number.
**Reason (R):** A composite number has more than two distinct positive factors. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

## SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

**Q7.** Find the HCF of 1260 and 7344 by prime factorisation. [2]

**Q8.** If the HCF of 65 and 117 is expressible in the form $65m - 117$, find the value of $m$. [2]

## SECTION C: Short Answer Type (2 × 3 = 6 Marks)

**Q9.** Show that any number of the form $8^n$, where $n \in \mathbb{N}$, can never end with the digit 0. [3]

**Q10.** Show that $15^n$ cannot end with the digit 0 for any natural number $n \in \mathbb{N}$. [3]

## SECTION D: Long Answer Type (2 × 5 = 10 Marks)

**Q11.** Prove that $\sqrt{2}$ is an irrational number. Hence, deduce that $\frac{5-3\sqrt{2}}{4}$ is an irrational number. [5]

**Q12.** Prove that $\sqrt{3}$ is an irrational number. Hence, prove that $\frac{1}{2+\sqrt{3}}$ is an irrational number. [5]

## SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

**Q13. CASE STUDY: CASE STUDY 07: AGRICULTURAL FRUIT ORCHARD HARVEST CRATE PACKING** [4 Marks]

A commercial fruit orchard in Himachal Pradesh harvests premium quality seasonal fruits. The daily harvest yields 144 kg of Royal Delicious apples, 180 kg of Kinnow oranges, and 240 kg of Bartlett pears. To minimize transport bruising, all fruits must be packed in identical wooden crates of maximum possible capacity such that each crate contains exactly one variety of fruit and no fruit is mixed.




**Crate Capacity** = **HCF**$(144, 180, 240) = 12$ **kg**

(i) Determine the maximum weight capacity of each crate. [2]

(ii) Calculate the total number of wooden crates required to pack the entire harvest. [1]

(iii) Find the value of $\text{HCF}(144, 180) \times \text{LCM}(144, 180)$. [1]

MATHS MANTHAN TUITION CLASSES